

4 Fig. 4.1 is a ribbon model of a molecule of haemoglobin.

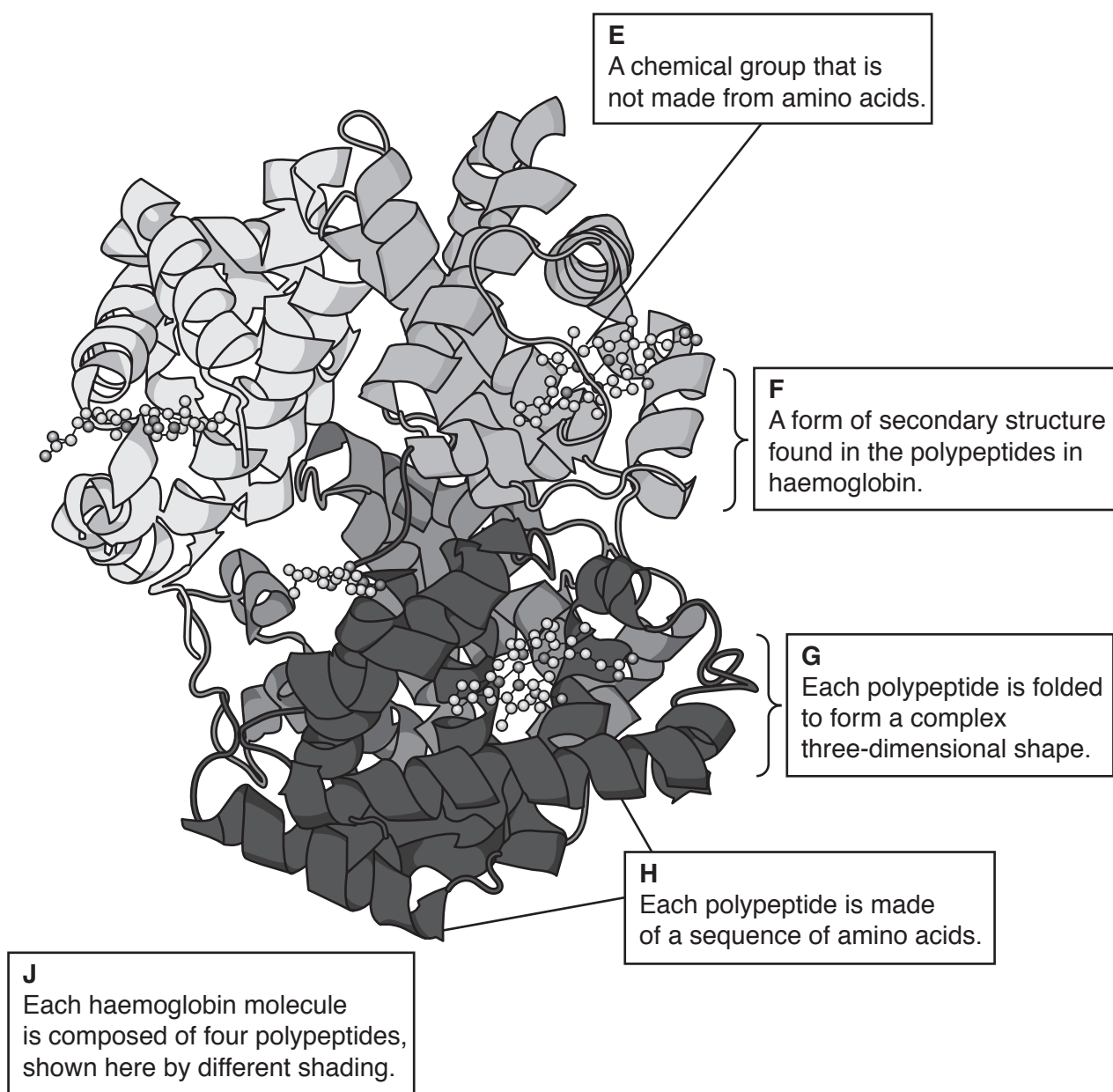


Fig. 4.1

(a) State the term that matches each of the descriptions given in boxes **E**, **F**, **G**, **H** and **J** in Fig. 4.1.

E

F

G

H

J

- (b) The effect of carbon dioxide on the percentage saturation of haemoglobin with oxygen was investigated.

A sample of mammalian blood was exposed to a gas mixture that contained increasing partial pressures of oxygen (pO_2). Throughout the investigation the partial pressure of carbon dioxide (pCO_2) was maintained at 5.3 kPa. The percentage saturation of haemoglobin in the sample was determined as the pO_2 increased.

The investigation was repeated with a second sample of blood in which the pCO_2 was maintained at 10.7 kPa.

The results are shown in Fig. 4.2.

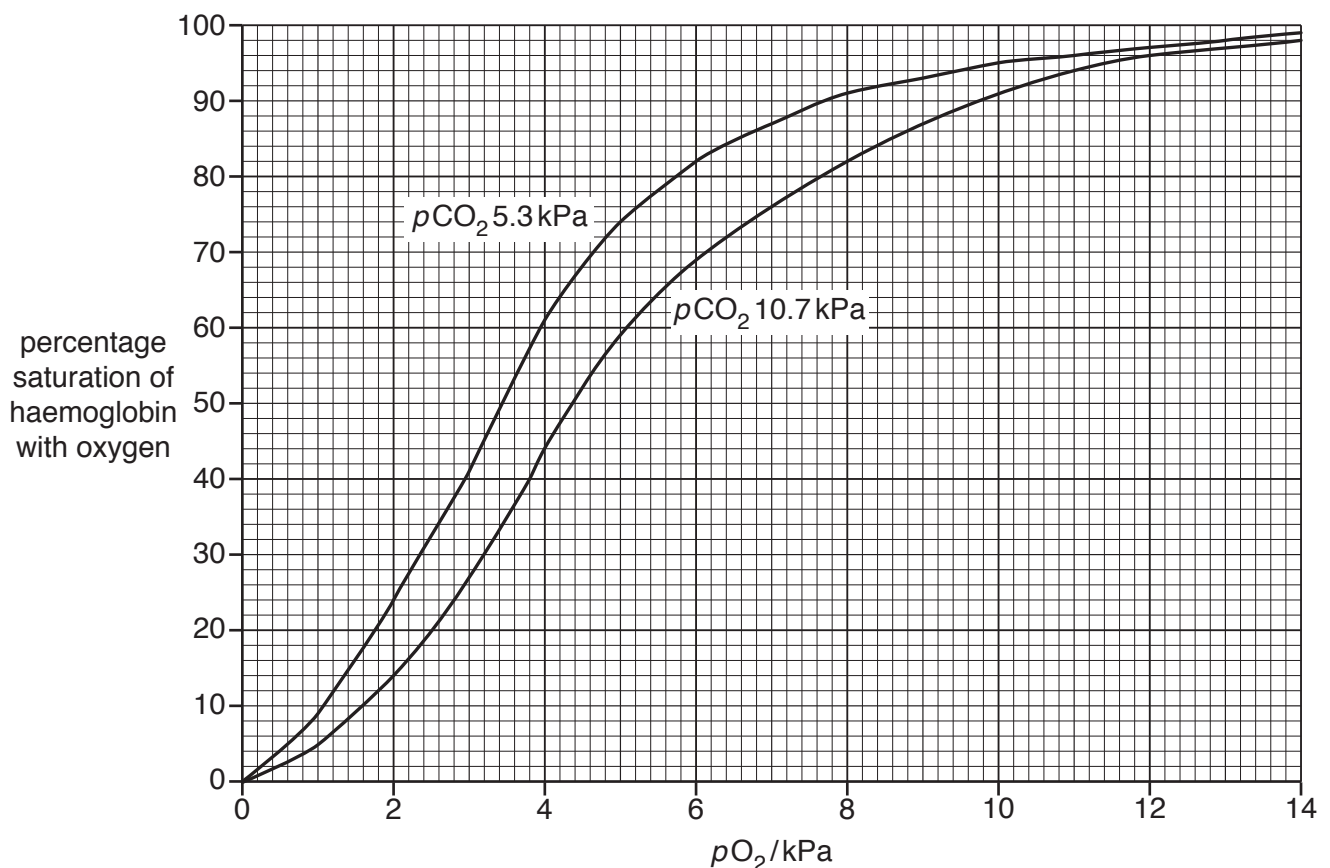


Fig. 4.2

- (i) The pO_2 in alveolar air is 13.0 kPa and the pCO_2 is 5.3 kPa.

Use Fig. 4.2 to suggest the percentage saturation of haemoglobin in blood entering the pulmonary veins.

.....[1]

- (ii) Use the information in Fig. 4.2 to state **and** explain the effect of increasing $p\text{CO}_2$ in muscle tissues during strenuous exercise.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 10]